

Pablo Aganza

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EXPERIENCE

Foresight Diagnostics

Remote

Data Engineer

July 2023 - January 2026

- Reduced scientific team dataset preparation and validation time from 7 days to under 1 hour by leading development of a scalable clinical data pipeline for FDA submission workflows
- Accelerated new clinical data source integration from 2-3 days of custom engineering work to under 2 hours of configuration by building reusable validation modules across varied file formats
- Cut data issue investigation from 3-5 days to under 4 hours by surfacing actionable quality failures through structured logging, validation reports, and auditable run records
- Saved 20+ software engineering hours per week by automating manual dataset generation support with self-service FDA submission-ready export tooling for Assay Development teams

Data Engineering Intern

January 2023 - July 2023

- Improved bioinformatics pipeline performance by performing threshold sensitivity analysis on diagnostic pipeline data

Genalyte

San Diego, CA

Machine Learning Engineer Intern

May 2022 - July 2022

- Automated blood diagnostic robot state classification by deploying a transfer-learning computer vision model to support R&D testing workflows and quality-control checks

PROJECTS

Imminent Seizure Prediction (Master's Degree Capstone)

Project Page

- Processed 572 GB of raw EEG seizure-monitoring data into a 75 GB model-ready dataset, creating labeled time-series sequences to predict seizures within the next 60 seconds
- Benchmarked supervised and unsupervised model families across patient-level evaluation splits, selecting an autoencoder approach based on the strongest overall classification performance
- Compared patient-specific and cross-patient modeling strategies, identifying individualized calibration as a key requirement for reliable seizure prediction across highly variable EEG patterns

Flight Departure Delay Prediction

- Built a Spark pipeline joining 700M records across flight, weather, and airport datasets to predict departure delays two hours before takeoff
- Engineered leakage-safe features from weather, airport congestion, carrier delay history, inbound aircraft movement, time patterns, and airport network structure to support realistic forecasting
- Selected XGBoost after benchmarking five model families and documenting trade-offs across predictive performance, interpretability, and operational complexity

EDUCATION

University of California, Berkeley

Berkeley, CA

M.S. Data Science; GPA: 3.92/4.00

January 2024 - May 2026

Carnegie Mellon University

Pittsburgh, PA

B.S. Mathematical Sciences; GPA: 3.30/4.00

August 2018 - May 2023

Minor in Computer Science, concentration in Robotics

TECHNICAL SKILLS

Big Data / Distributed: Apache Spark, PySpark, Spark SQL, MLlib, Databricks, Hadoop, MapReduce, MrJob

Data Platforms: PostgreSQL, BigQuery, Neo4j, Redis

Data Science Libraries: Pandas, NumPy, Matplotlib, Seaborn

Languages: Python, SQL, R, Bash

Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, Hugging Face Transformers

ML Tooling / MLOps: Optuna, FastAPI, Pydantic

Platforms / DevOps: GCP, AWS, Docker, Kubernetes, Git, GitHub Actions, CI/CD